

$$\begin{aligned} 3a. \quad \Delta y &= \frac{\lambda D}{d} \\ &= \frac{vD}{fd} \end{aligned}$$

$$\begin{aligned} \text{Frequency of sound, } f &= \frac{vD}{d\Delta y} \\ &= \frac{340 \times 12}{2.5 \times 0.60} \\ &= 2720 \text{ Hz} \end{aligned}$$

$$\text{bi.} \quad 2A$$

$$\text{ii.} \quad \frac{A}{\sqrt{2}} = 0.707 A$$

$$\begin{aligned} \text{iii.} \quad \left( A + \frac{A}{\sqrt{2}} \right) &= 1.7071A \\ &\approx 1.71A \end{aligned}$$

$$\begin{aligned} \text{iv.} \quad \text{Old intensity: } I &= k(2A)^2, \quad k = \text{constant} \\ &= 4kA^2 \end{aligned}$$

$$\begin{aligned} \text{New intensity: } I' &= k(1.7071A)^2 \\ &= 2.91kA^2 \\ &= 0.729 I \end{aligned}$$